

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence (AI) and Data Science) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Artificial Intelligence (AI) and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTAIC602 - ADVANCED MACHINE LEARNING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 In K-Means clustering, the objective function minimizes:
 - A. Inter-cluster distance
 - B. Intra-cluster variance
 - C. Number of clusters
 - D. Distance between medoids
 - 2 The Silhouette score measures:
 - A. Cluster size
 - B. Classification accuracy
 - C. How similar an object is to its own cluster compared to other clusters
 - D. Model complexity
 - 3 Which parameter measures how often an itemset appears in the dataset?
 - A. Confidence
 - B. Lift
 - C. Support
 - D. Conviction
 - 4 Collaborative filtering primarily relies on:
 - A. Item features
 - B. User behavior and preferences
 - C. Expert rules
 - D. Market basket data
 - 5 The agent–environment interaction in reinforcement learning is formalized using:
 - A. Bayesian Network
 - B. Decision Tree
 - C. Markov Decision Process
 - D. Neural Network
 - 6 In ϵ -greedy strategy, ϵ represents:
 - A. Learning rate
 - B. Discount factor
 - C. Probability of random action
 - D. Reward function

- 7 Which of the following is NOT a component of time series?
 - A. Trend
 - B. Seasonality
 - C. Noise
 - D. Clustering
- 8 A time series is a sequence of data points indexed by:
 - A. Frequency
 - B. Time
 - C. Probability
 - D. Features
- 9 Model selection aims to choose a model that:
 - A. Fits training data perfectly
 - B. Minimizes generalization error
 - C. Has maximum parameters
 - D. Runs fastest
- 10 Cross-validation helps in:
 - A. Increasing training data
 - B. Reducing bias and variance
 - C. Feature extraction
 - D. Data normalization
- 11 Adaptive Boosting is commonly known as:
 - A. Gradient Boosting
 - B. XGBoost
 - C. AdaBoost
 - D. Bagging
- 12 ARIMA stands for:
 - A. Auto Regression Integrated Moving Average
 - B. Advanced Regression Integrated Model Algorithm
 - C. Auto Random Integrated Moving Algorithm
 - D. Adaptive Regression Integrated Model

Q2. Solve the following.

- A) Given the following 2D data points: 6
 (2,2), (4,4), (6,6), (8,8) and $K = 2$
 a) Initialize centroids at (2,2) and (8,8)
 b) Perform two iterations of K-Means and compute the updated centroids.
- B) Explain the Silhouette method for selecting optimal K. Why is it preferred over the elbow method in some cases? 6

Q3. Solve the following.

- A) Define association rule mining and explain the terms 'support,' 'confidence,' and 'lift.' 6
- B) Differentiate between collaborative filtering and content-based filtering with suitable examples. 6

Q4. Solve Any Two of the following.

- A) Given the following Q-Learning parameters: 6
 $Q(s,a)=2$, reward = 5, learning rate $\alpha = 0.5$, discount factor $\gamma = 0.9$, $\max Q(s')=4$
 Calculate the **updated Q-value** using the Q-learning update rule.
- B) Compare Q-Learning and SARSA algorithms in terms of policy learning and convergence. 6

- C) Design a reinforcement learning model for traffic signal control and explain states, actions, rewards, and policy. 6
- Q5. Solve Any Two of the following.
- A) Given a time series with mean = 50 and variance = 25, determine whether it can be considered stationary. Justify your answer. 6
- B) Analyze the differences between AR, MA, and ARMA models with equations. 6
- C) Explain how ACF and PACF plots help in selecting ARIMA model parameters. 6
- Q6. Solve Any Two of the following.
- A) A model produces the following accuracies in 5-fold cross-validation: 78%, 80%, 82%, 79%, 81% Compute the average accuracy and comment on model stability. 6
- B) Analyze the working of boosting algorithms with emphasis on weak learners. 6
- C) Explain cross-validation and its role in avoiding overfitting. 6

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